

DATA VISUALIZATION:

Syllabus:

Expt 01: Finalizing dataset

Expt 02: Data Pre-processing

Expt 03: Importing and exporting data

Expt 04: Graphs

Link:  DV Honors

Sample dataset:

<https://www.kaggle.com/datasets/nelqiriyewithana/indian-weather-repository-daily-snapshot?source=download>

1. Preprocessing

(updating Excel formulae and ways of preprocessing as they are found.)

(I) To remove duplicates:

Go to Data -> Remove Duplicates

(II) Uppercase/lowercase standardized rows:

Create a new column ->

UPPER (CELL_number) (all letters uppercase)

PROPER (CELL_number) (first letter in name, uppercase)

LOWER (CELL_number) (all letters lowercase)

(III) Use a filter in the columns and find spelling mistakes. Or garbage values. Modify them..

(IV) To remove extra spaces:

TRIM (CELL_number)

(V) If numbers aren't being imported properly and the symbol in it causes a problem:

Go to Home -> In Number -> Change Data Type

(VI) To format dates to look in a standard format:

Go to Home -> In Number -> Change Data Type to Short Date

(*) If pasting a column over another, after processing; paste as values. Second option while pasting.

(*) If any column is redundant, delete it completely.

(VII) Counting non-empty cells in a row:

=COUNTA (A1:A10)

Counting empty cells in a row: =COUNTBLANK (A1:A10)

(If more than 75% of the cells in a row are empty, row can be deleted)

(VIII) To fill missing numerical data:

In the beginning, select the cell range as they are the first and last cells between the missing values. Go to the Home Tab and click the Fill icon. Select Series from Drop Down Menu. In the Series window, select Columns and Growth under the Series in and Type sections respectively. Mark the Trend box. Press Ok.

2. Graphs and Other Concepts:

- A. Pie chart
 - B. Donut chart
 - C. Lollipop chart
 - D. Boxplot
 - E. Bubble chart
 - F. Bump Chart
 - G. Funnel chart
 - H. Water Chart
 - I. Bar chart /Column chart /Stacked chart
 - J. Area Chart
 - K. Pareto chart
 - L. Tree maps
 - M. Scatter plot
 - N. Gantt chart
 - O. World map
-
- P. Parameters
 - Q. Combine, Sets
 - R. Label, Title and caption
 - S. Working with metadata
 - T. Working with filter
 - U. Font, Size, Color, and Border,
 - V. Calculated field, Number function, string function
 - W. Conditional formatting, logical function
 - X. Exporting the charts

3. Theory

(ppt links and previous papers)

https://drive.google.com/drive/folders/1v6HTCLM-tuB2pAT6GTN0rJWDtzLR3PdO?usp=drive_li nk

Books on Tableau/Data Visualization:

- [sosulski-kristen-data-visualization-made-simple_-insights-into-becoming-visual-2019-routl...](#)
- [dokumen.pub_visual-data-storytelling-with-tableau-1ed-.pdf](#)
- [storytelling-with-data-cole-nussbaumer-knaflic.pdf](#)